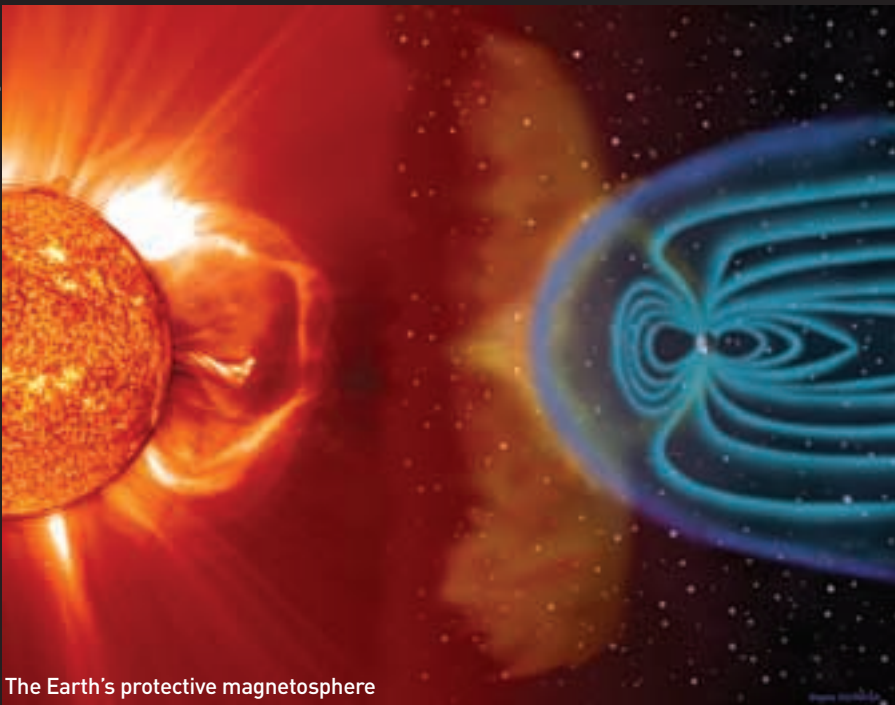


Your devices are doomed... ...and you may be too!

Image: NASA



The Earth's protective magnetosphere

Thanks to Hollywood, all of us now know that the Mayans believed that the world would end in 2012. Of course we may have got it all wrong, and Mayan priests and historians who actually speak the language translate the whole thing into a “refreshing new start”, but that’s not as exciting as Armageddon, is it? It takes a lot more to destroy a planet than it does to kill humanity, and to paraphrase from a famous documentary, billions of years from now the Earth will still be here, mankind won’t.

RA – GOD OR GOON?

The ancient Egyptians believed that the sun was a god, the creator. The sun meant light and warmth, which was considered essential for survival. Of course, even today we know that without our sun our planet would be just a hunk of cold rock and ice, and so the title of creator might actually be fitting.

However, he who creates can also destroy!

SPACE CLIMATE

Most people consider space to be a lifeless, absolute-zero void, so to consider that space actually has a climate seems far-fetched. The truth is, within our solar system – as it is within almost every one of the trillions of solar systems out there – there is a very active climate system. The sun isn’t just a super-massive ball of burning gas; it also has storms, eruptions and as some would put it, mood swings. It’s almost alive, and temperamental, and can cause us a lot of harm when it’s angry!

Solar weather is caused by the intense heat that makes the sun what it is. At these temperatures, gasses such as Hydrogen and Helium are broken down into ions (electrons and protons), and form plasma – the fourth state of matter after the more popular solid, liquid and gaseous states.



Makes you feel miniscule, doesn't it?

Although plasma is basically a gas with ionised particles, unlike gasses, it is very prone to magnetic effects, and this causes a massive magnetic system on the sun.

Just like our weather, the solar weather has a cycle, and it’s about 11 years long. From periods of increased activity and clearly visible sun spots to lulls and then back again takes 11 years. Since the solar cycle is magnetic based, the sun actually reverses its magnetic poles once every such cycle. Thus the magnetic cycle (north to south and back again) actually takes 22 years, but since no

WATCH THE SUN IN ACTION

Make sure to visit <http://bit.ly/VbgUU>, and check out the way scientists are tracking solar flares and coronal mass ejections. You can select a time frame and watch the sun’s activities at fast speeds. And remember, never look at the sun with the naked eye or you could do some damage..

one cares which way a compass points on the surface of the sun, and we’re all a little more concerned about solar flares and the like, 11 years is the accepted climate cycle that everyone remembers.

SUNNY DAYS

Sun spots are basically the areas in which the largest magnetic storms rage on the sun.

GEOMAGNETIC STORMS

When solar events occur, radiation and ion particles released from the sun can cause interference with the earth’s magnetic field. When a large such event happens, the Earth can experience an electromotive force, which can knock out technology such as satellites and power grids.

SOLAR SUPER-STORM OF 1859

The only recorded solar superstorm in known history, this CME event was so large that aurora borealis were seen as far south as the Caribbean! There were also some reports of miners in the Rocky mountains waking up in the middle of the night to fix breakfast because they thought the Aurora Borealis was early morning sunlight!